bank with a higher response time, as far as you are concerned. Thus, Flash storage definitely improves customer experience. From the customer's investment point of view, Flash is all about saving on the power, energy and data centre footprint. We have real examples where customers have actually reduced their data centre footprint by almost three times and saved on the power by almost seven times—just by using Flash. So, these are the compelling reasons why I feel Flash is becoming increasingly popular.

Q When a CIO has already invested the firm's CAPEX in a storage solution provided by you, how can the company leverage the best features of this new storage solution going forward?

Being the storage head of HP India, I get information about the upcoming products at least three to four months prior to their launch. But even though I have this information, I cannot share anything with a customer till the date of the product launch. It's a policy related matter. Even if I want to share any information with a customer before the official launch date, I will have to take special permissions. But in the special cases of enterprise customers with whom we have very large infrastructure deployments, we sign a NDA and share the specifications of the upcoming products. This is to ensure that they can use this information when they come up with their requirements. But this is done on a case-to-case basis and not on a mass scale. For clients who have been with HP for a long time, like a period of 10 years or so, any upgradation is called a technology refresh. For other clients, as per our 'embedded lease' offering, HP will have to replace the old infrastructure and also take the responsibility of migrating from the old storage architecture to the new one.

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Q What are the current challenges and trends in the storage industry?

Customers who move from a hybrid landscape or a spinning landscape to Flash need to get ready for that first. It is critical that they train their human resources in how to utilise Flash in an optimised manner. If Flash is not properly used, or is left under-utilised, then clients can actually end up paying more for less work. So, they should be sensitive to that aspect and gear up the organisation for Flash usage.

So, when we are selling these all-Flash array solutions to customers, we are typically bundling in three days of training on how to use Flash architecture, because we feel that is an important aspect as it is a mega transition.

When we talk about the trends that are picking up in the storage industry, software defined storage is definitely one of them. All-Flash arrays are another trend that is picking up strongly. The third most important trend is disk-to-disk backup instead of tape backup.

Q How have things changed for the storage industry in the past one year, particularly with software defined infrastructure coming in?

Software defined storage (SDS) is a very small segment in India right now, but there is no doubt that it will continue to grow. But if you ask me whether it will reach the double digit growth in India in the near future, my answer is 'No'. However, we cannot deny the fact that it is very cost-effective and extremely easy to deploy. We have easily deployed our SDS on a network within hours. With all these factors, SDS is poised to grow. But if customers are looking for performance, then they need to look at Flash based architectures.

Q How is software defined storage faring in the current computing scenario?

Let me answer this question with a small example of SDS, which is a performance based solution. If customers are looking at the lowest cost with the fastest deployment, they should look at SDS. Let's take the example of a small company that has 10 branches. The IT head of the company has got 10 servers and each of the servers has 300 X 2GB of disk space. Assuming that location #3 has run out of disk space and the requirement is 700GB, the IT head will have to buy the additional 100GB of space. In case of an SDS deployment, the IT head will deploy a licence on the headquarters' server and appellate on the remaining nine servers. While deploying SDS, the IT head of this organisation can see the status of the 6000GB of space, and will not need to spend a single penny till that space has been fully allocated to all locations, allowing the excess storage space at other locations to be assigned to location #3.

So, it goes back to the philosophy of converged infrastructure, where I am able to get better optimisation because of an SDS licence. In a nutshell, with SDS, the resources of all attached devices can be converted into a resource pool. And an IT head can get a full view of that storage, which will eventually prove to be a very cost-effective option. It may not give you performance, but will give you optimisation and efficiency.